Lab Experiments:

Set 6

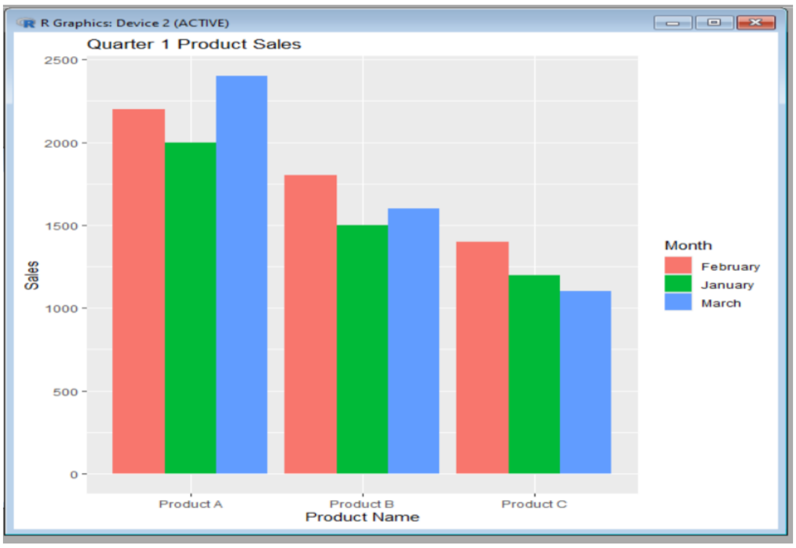
1. **Grouped bar chart to visualize the sales of each product for the first quarter:**

Python code:

```
sales <- data.frame(
 ProductID = c(1,2,3),
 ProductName = c("Product A","Product B","Product C"),
 January = c(2000,1500,1200),
 February = c(2200,1800,1400),
 March = c(2400,1600,1100)
)
sales_long <- pivot_longer(
 sales,
 cols = c(January, February, March),
 names_to = "Month",
 values_to = "Sales"
)

ggplot(sales_long,
     aes(x = ProductName,
         y = Sales,
         fill = Month)) +
 geom_bar(stat = "identity",
         position = "dodge") +
 labs(title = "Quarter 1 Product Sales",
     x = "Product Name",
     y = "Sales")
```

Output:

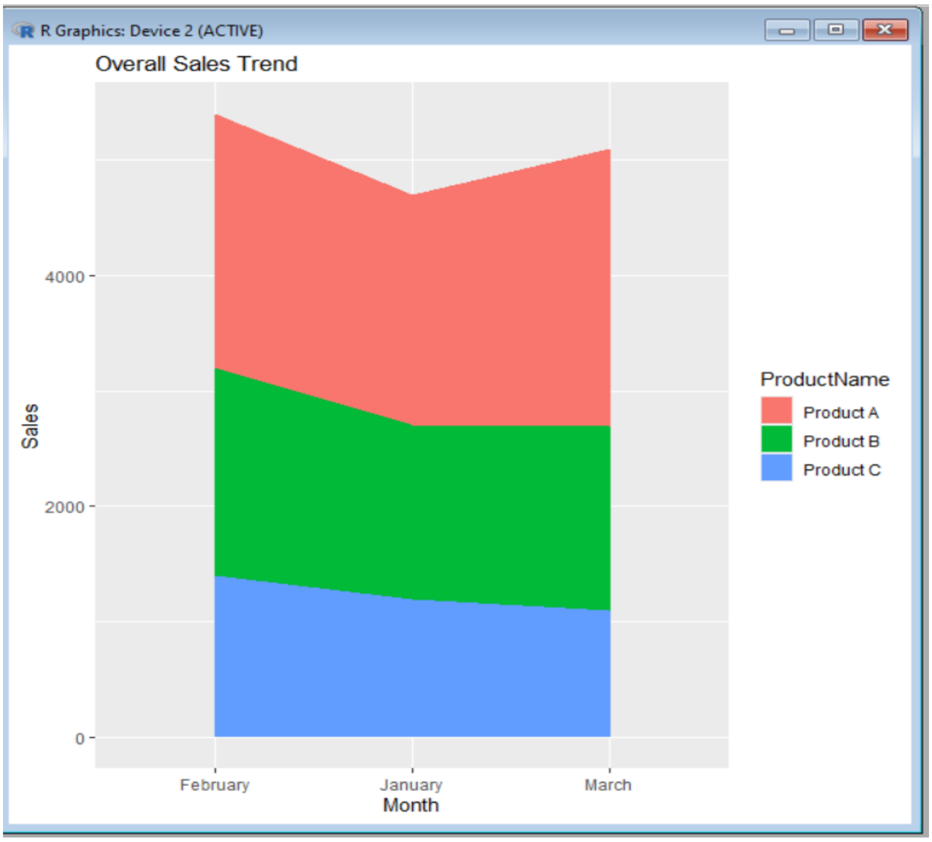


2. **Stacked area chart to represent the overall sales trend for all products over the three months:**

Python code:

```
sales <- data.frame(
 ProductID = c(1,2,3),
 ProductName = c("Product A","Product B","Product C"),
 January = c(2000,1500,1200),
 February = c(2200,1800,1400),
 March = c(2400,1600,1100)
)
ggplot(sales_long,
    aes(x = Month,
      y = Sales,
      fill = ProductName,
      group = ProductName)) +
 geom_area() +
 labs(title = "Overall Sales Trend",
    x = "Month",
    y = "Sales")
```

Output:



3. **Show the monthly sales data for each product:**

   Code:

   grid.table(sales)

   Output:

	ProductID	ProductName	January	February	March
1	1	Product A	2000	2200	2400
2	2	Product B	1500	1800	1600
3	3	Product C	1200	1400	1100

4. **The grouped bar chart, stacked area chart, and the table for interactive exploration of sales data:**

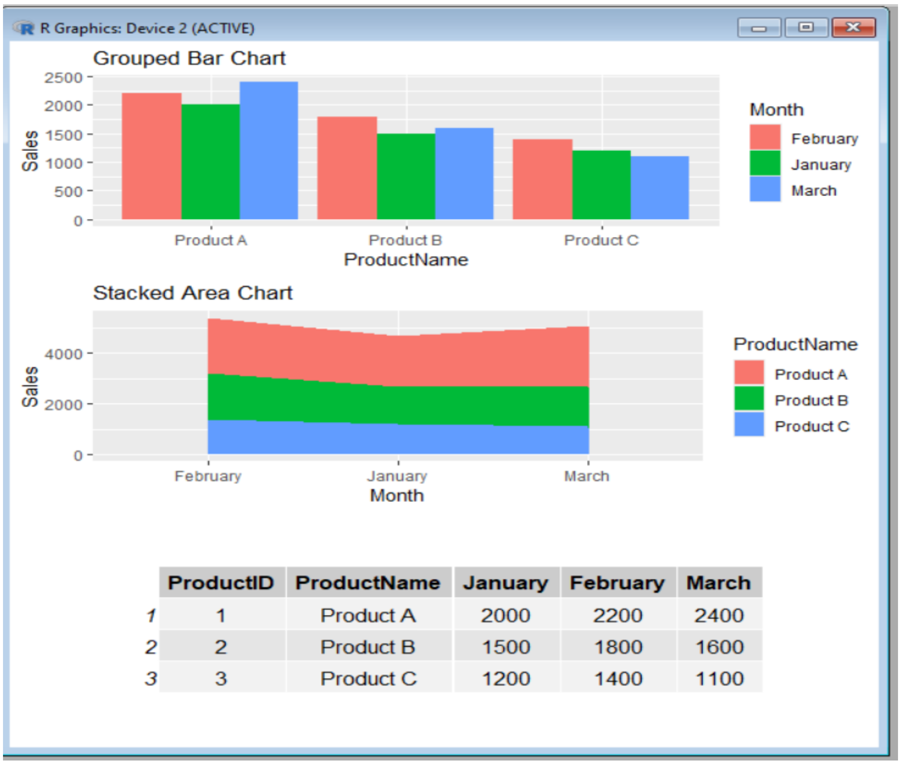
   Python code:

```
p1 <- ggplot(sales_long,
        aes(ProductName, Sales, fill = Month)) +
   geom_bar(stat = "identity",
         position = "dodge") +
   labs(title = "Grouped Bar Chart")

p2 <- ggplot(sales_long,
        aes(Month, Sales,
          fill = ProductName,
          group = ProductName)) +
   geom_area() +
   labs(title = "Stacked Area Chart")

grid.arrange(
 p1,
 p2,
 tableGrob(sales),
 ncol = 1
)
```

Output:



	ProductID	ProductName	January	February	March
1	1	Product A	2000	2200	2400
2	2	Product B	1500	1800	1600
3	3	Product C	1200	1400	1100

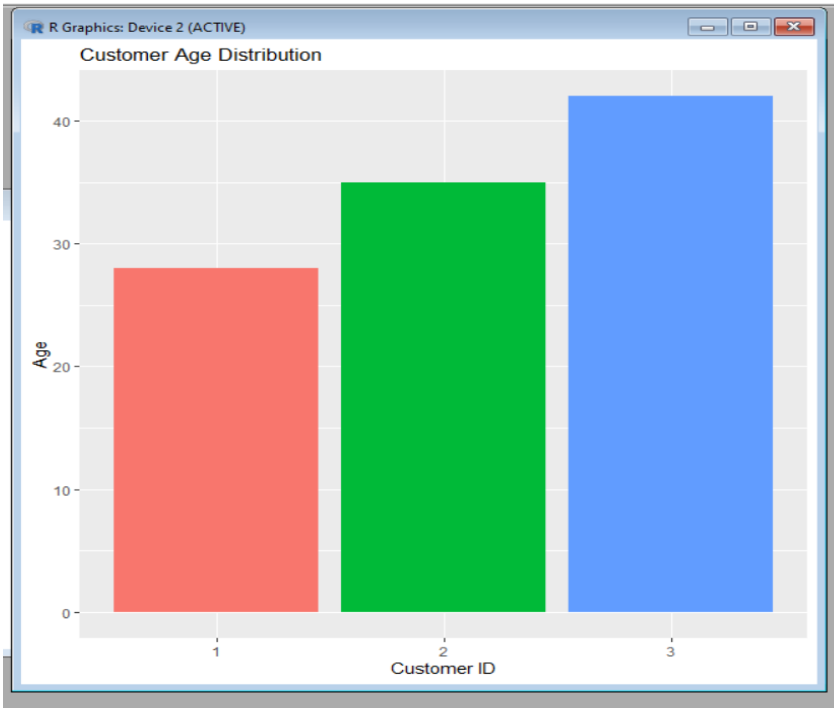
Set 7

1. **bar chart to visualize the distribution of customer ages:**

   python code:

```
customer <- data.frame(
 CustomerID = c(1,2,3),
 Age = c(28,35,42),
 Gender = c("Female","Male","Female"),
 Income = c(50000,60000,75000)
)
ggplot(customer,
    aes(x = factor(CustomerID),
      y = Age,
      fill = factor(CustomerID))) +
 geom_bar(stat = "identity") +
 labs(title = "Customer Age Distribution",
    x = "Customer ID",
    y = "Age") +
 theme(legend.position = "none")
```

Output:

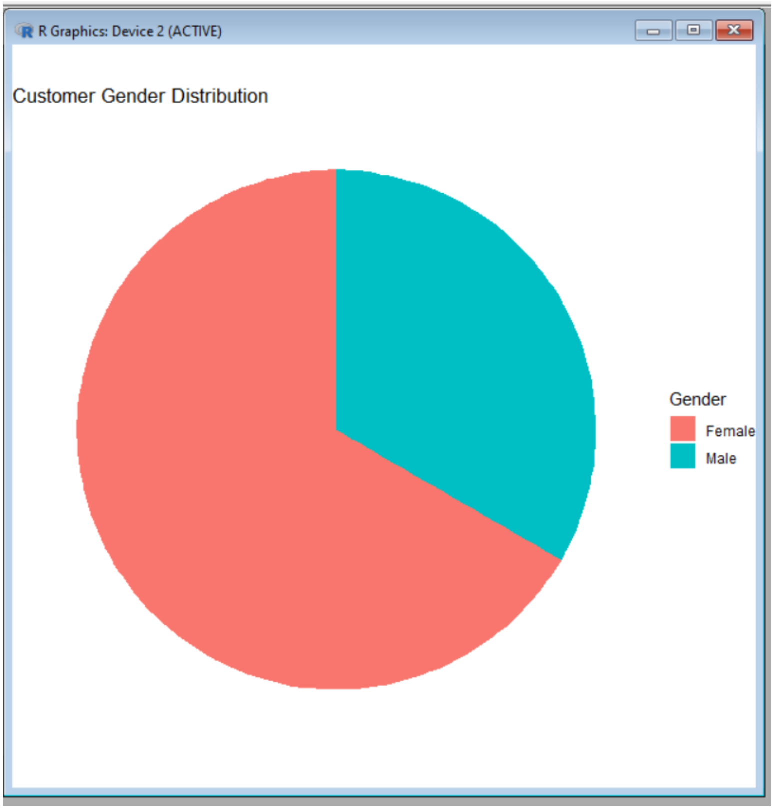


2. **pie chart to represent the distribution of customers by gender:**
   python code:

```
customer <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
gender_data <- as.data.frame(table(customer$Gender))
colnames(gender_data) <- c("Gender", "Count")

ggplot(gender_data,
     aes(x = "", y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(title = "Customer Gender Distribution") +
  theme_void()
```

Output:



3. **table to show the demographic information for each customer.**

   Python code:

   grid.table(customer)

   Output:

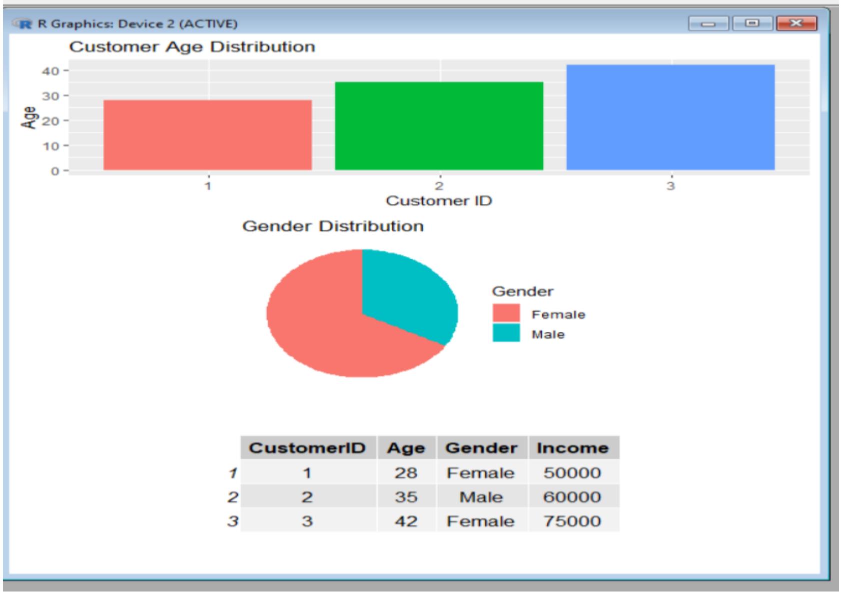
	CustomerID	Age	Gender	Income
1	1	28	Female	50000
2	2	35	Male	60000
3	3	42	Female	75000

4. **combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.**

Python code:

```
p1 <- ggplot(customer,
          aes(x = factor(CustomerID),
            y = Age,
            fill = factor(CustomerID))) +
 geom_bar(stat = "identity") +
 labs(title = "Customer Age Distribution",
    x = "Customer ID",
    y = "Age") +
 theme(legend.position = "none")
gender_data <- as.data.frame(table(customer$Gender))
colnames(gender_data) <- c("Gender", "Count")
p2 <- ggplot(gender_data,
          aes(x = "", y = Count, fill = Gender)) +
 geom_bar(stat = "identity", width = 1) +
 coord_polar("y") +
 labs(title = "Gender Distribution") +
 theme_void()
grid.arrange(
 p1,
 p2,
 tableGrob(customer),
 ncol = 1
)
```

Output:



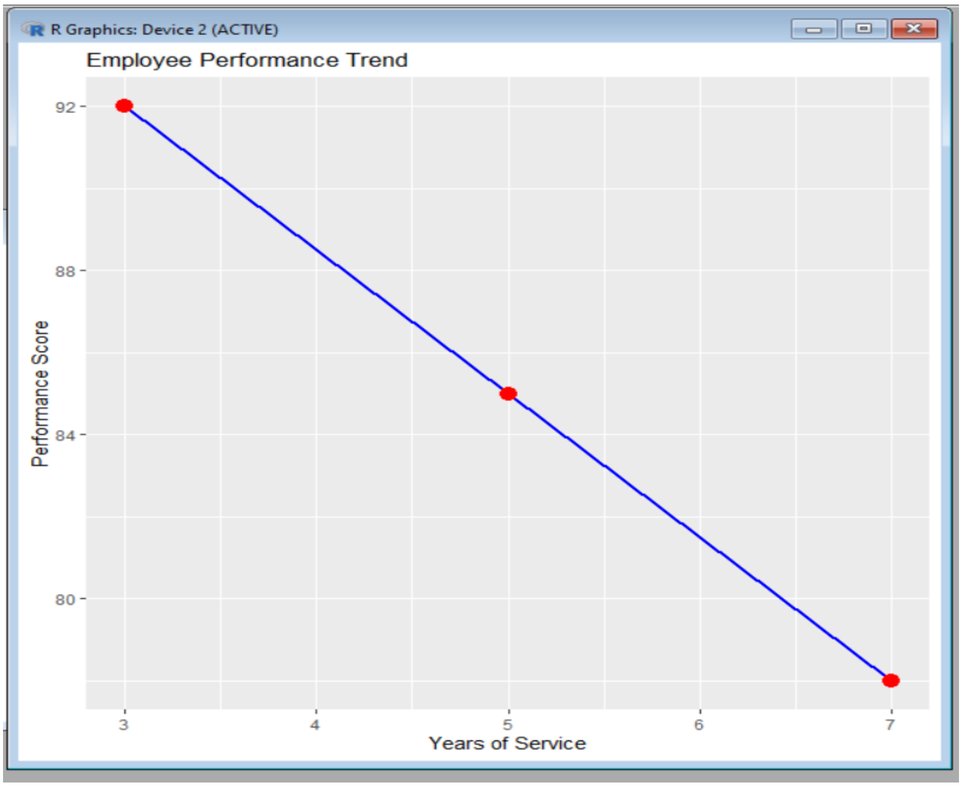
Set 8

1. **line chart to visualize the performance trend of employees over time.**

Python code:

```
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
ggplot(employee,
       aes(x = YearsService,
           y = PerformanceScore,
           group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(size = 4, color = "red") +
  labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score"
  )
```

Output:

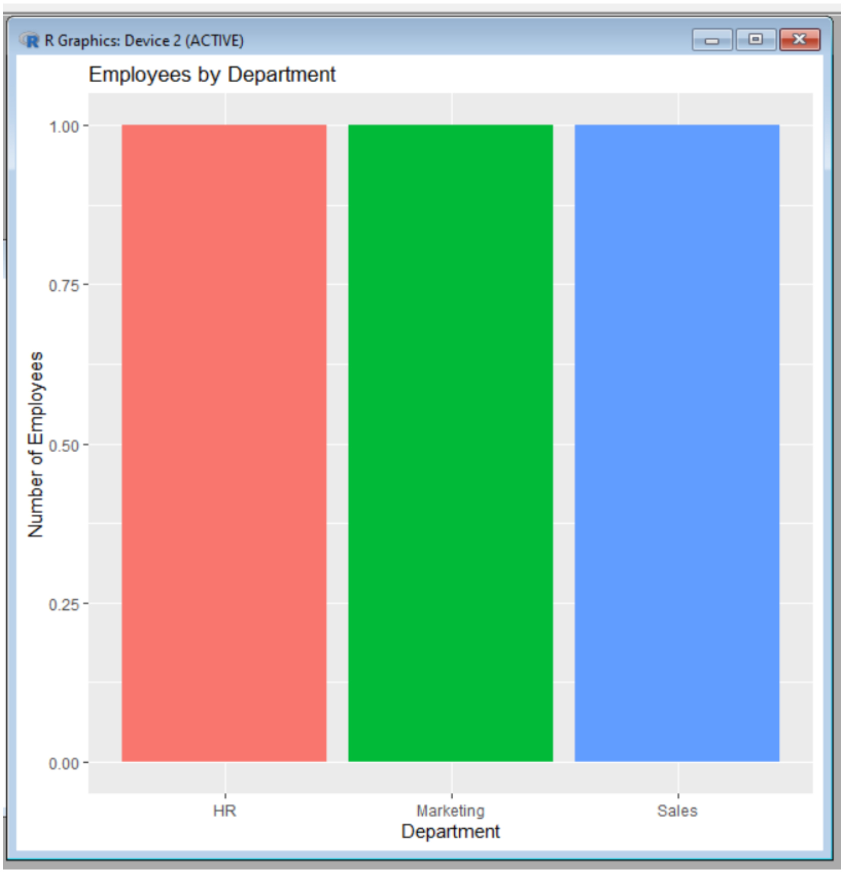


2. **bar chart showing the distribution of employees across different departments:**

python code:

```
employee <- data.frame(
 EmployeeID = c(1,2,3),
 Department = c("Sales","HR","Marketing"),
 YearsService = c(5,3,7),
 PerformanceScore = c(85,92,78)
)
ggplot(employee,
     aes(x = Department,
         fill = Department)) +
 geom_bar() +
 labs(
  title = "Employees by Department",
  x = "Department",
  y = "Number of Employees"
 ) +
 theme(legend.position = "none")
```

Output:



3. **table to display the performance data for each employee:**

   python code:

```
grid.table(employee)
```

   output:

	EmployeeID	Department	YearsService	PerformanceScore
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

4. **combining the line chart, bar chart, and the table for interactive exploration of employee performance data:**

```
library(gridExtra)

# Line Chart
p1 <- ggplot(employee,
         aes(x = YearsService,
           y = PerformanceScore,
           group = 1)) +
   geom_line(color = "blue", linewidth = 1) +
   geom_point(size = 4, color = "red") +
   labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score"
   )

# Bar Chart
p2 <- ggplot(employee,
         aes(x = Department,
           fill = Department)) +
   geom_bar() +
   labs(
    title = "Employees by Department",
    x = "Department",
    y = "Count"
   ) +
   theme(legend.position = "none")
```

```
# Dashboard
grid.arrange(
  p1,
  p2,
  tableGrob(employee),
  ncol = 1
)
```

Output:

	EmployeeID	Department	YearsService	PerformanceScore
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

Set 9

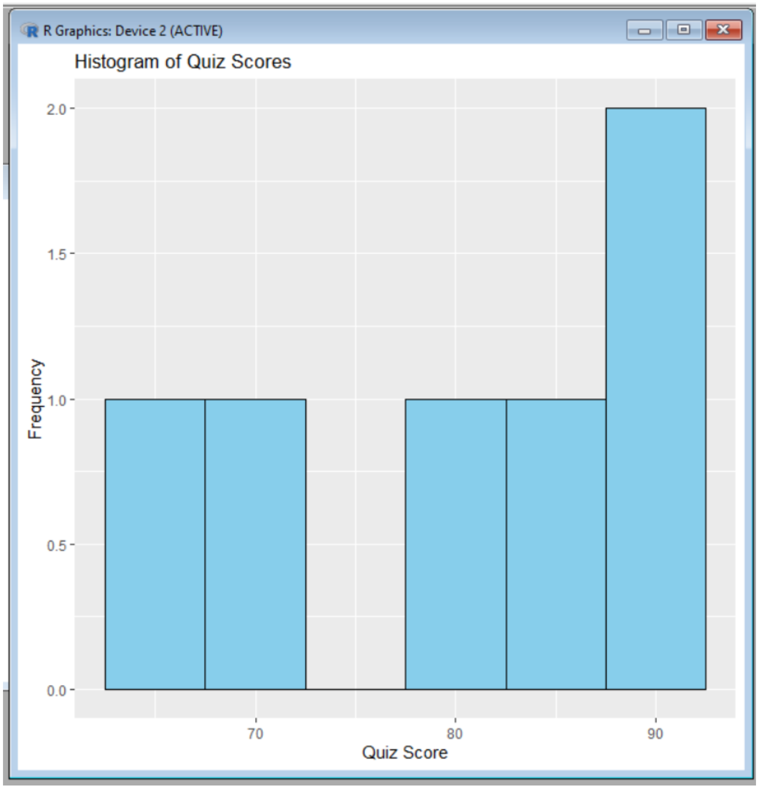
1. **histogram for Quiz_Score & boxplot of Quiz_Score by Course.**

   1.a. Histogram:

   Python code:

```
ggplot(data, aes(x = Quiz_Score)) +
  geom_histogram(binwidth = 5,
                 fill = "skyblue",
                 color = "black") +
  labs(
    title = "Histogram of Quiz Scores",
    x = "Quiz Score",
    y = "Frequency"
  )
```

Output:

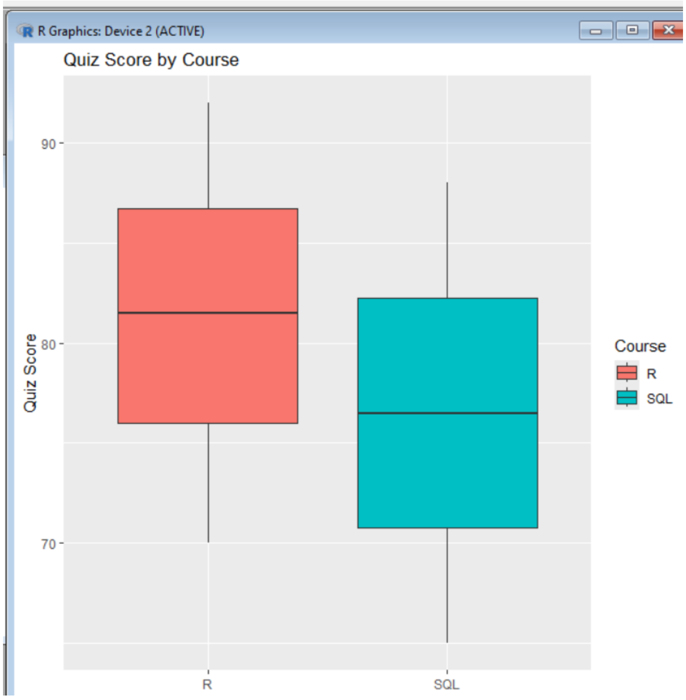


1.b.Boxplot:

Python code:

```
ggplot(data,
      aes(x = Course,
          y = Quiz_Score,
          fill = Course)) +
 geom_boxplot() +
 labs(
   title = "Quiz Score by Course",
   x = "Course",
   y = "Quiz Score"
 )
```

Output:

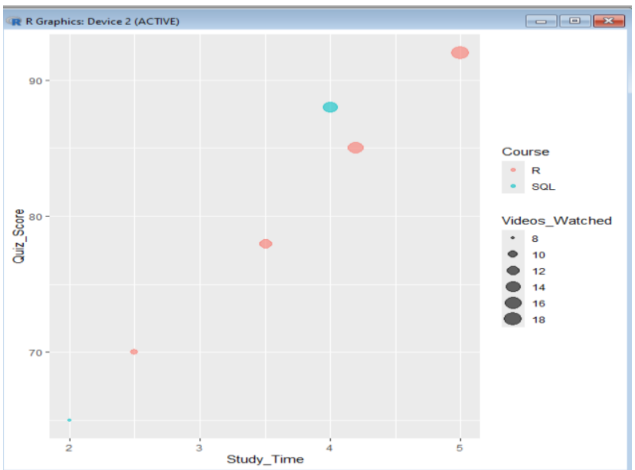


2. **between Study_Time and Quiz_Score.:**

python code:

```
ggplot(data,
      aes(x = Study_Time,
          y = Quiz_Score,
          size = Videos_Watched,
          color = Course)) +
 geom_point(alpha = 0.6) +
 labs(
   title = "Study Time vs Quiz Score",
   x = "Study Time (hrs)",
   y = "Quiz Score"
 )
```

Output:



3. **convert Login_Date into Date format.**
   Python code:

```
ggplot(monthly,
      aes(x = Month,
          y = Average_Score,
          group = 1)) +
 geom_line(color = "blue", linewidth = 1) +
 geom_point(size = 3) +
 geom_line(aes(y = MovingAverage),
           color = "red",
           linetype = "dashed",
           linewidth = 1) +
 labs(
  title = "Average Monthly Quiz Score",
  x = "Month",
  y = "Average Quiz Score"
 )
```

Output:

Set 10

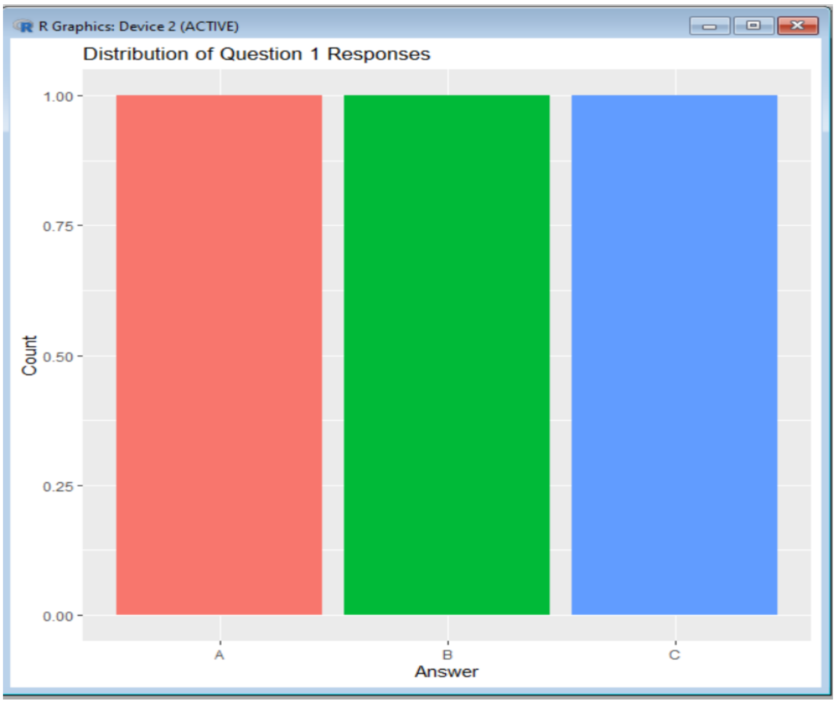
1. **grouped bar chart to visualize the distribution**

python code:

```
survey <- data.frame(
 SurveyID = c(1,2,3),
 Question1 = c("A","B","C"),
 Question2 = c("B","A","A"),
 Question3 = c("C","D","B")
)

survey
ggplot(survey,
    aes(x = Question1,
      fill = Question1)) +
 geom_bar() +
 labs(
  title = "Distribution of Question 1 Responses",
  x = "Answer",
  y = "Count"
 ) +
 theme(legend.position = "none")
```

Output:



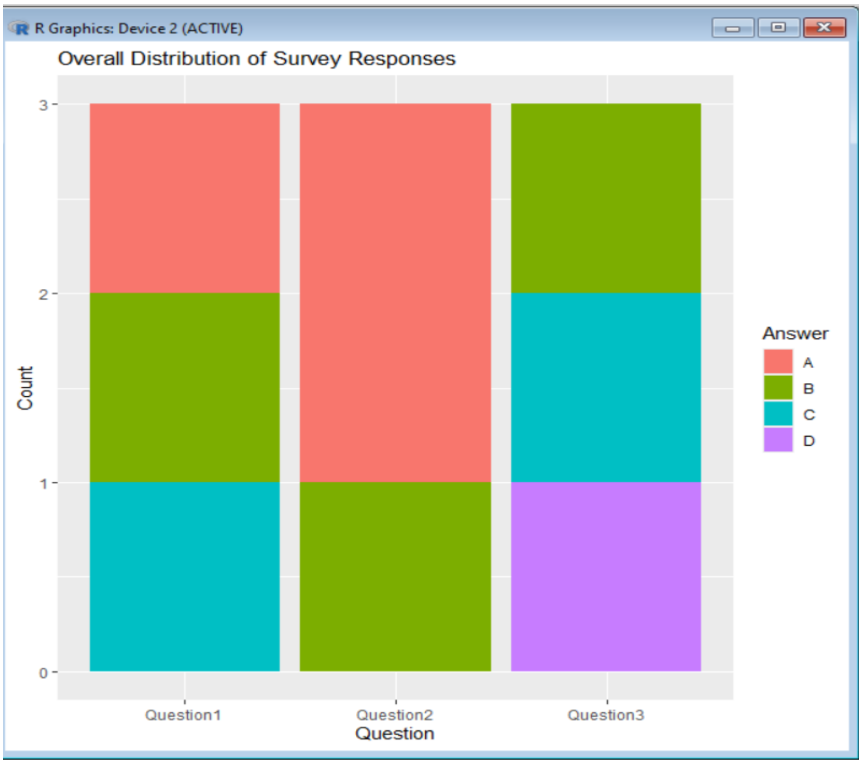
2. **stacked bar chart to represent the overall distribution of responses**

python code:

```
survey_long <- pivot_longer(
  survey,
  cols = c(Question1, Question2, Question3),
  names_to = "Question",
  values_to = "Answer"
)

ggplot(survey_long,
       aes(x = Question,
           fill = Answer)) +
  geom_bar() +
  labs(
    title = "Overall Distribution of Survey Responses",
    x = "Question",
    y = "Count"
  )
```

Output:



3. **table to show the survey response data for each survey.**

Python code:

```
grid.table(survey)
```

Output:

	SurveyID	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

4. **combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.**

Python code:

```
p1 <- ggplot(survey,
             aes(x = Question1,
                 fill = Question1)) +
  geom_bar() +
  labs(
    title = "Question 1 Responses",
    x = "Answer",
    y = "Count"
  ) +
  theme(legend.position = "none")
survey_long <- pivot_longer(
  survey,
  cols = c(Question1, Question2, Question3),
  names_to = "Question",
  values_to = "Answer"
)

p2 <- ggplot(survey_long,
             aes(x = Question,
                 fill = Answer)) +
  geom_bar() +
  labs(
    title = "Overall Survey Responses",
    x = "Question",
    y = "Count"
  )
```

```
grid.arrange(
  p1,
  p2,
  tableGrob(survey),
  ncol = 1
)
```

Output:

	SurveyID	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B